PROJECT #1 · WEB APPLICATION PROJECT PROPOSAL

Perde

An event discovery and ticketing web application

PROJECT TITLE	Perde — Event Discovery & Ticketing Platform
GROUP NUMBER	Not assigned
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COURSE	SOFT3111.1 – Internet and Web Programming Instructor: Mustafa Taner Eskil
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GITHUB REPOSITORY	github.com/kedroo/SOFT3111-Perde

“Perde” is Turkish for “curtain”. The repository also contains the screen sketches (Figma file and PDF) and a navigable HTML prototype of the user interface.

1 Project Overview and Motivation

What the application is about. Perde is a web application for discovering live events and buying tickets in Istanbul, Ankara and Izmir. It brings concerts, theatre, stand-up comedy, cinema, sports and children's shows into one catalogue and takes the user from "what should we do this weekend?" to an e-ticket with a QR code in a few steps.

The problem. Buying a ticket online is often more stressful than it should be. Popular platforms present long lists that can only be searched by keyword, so subjective needs such as "something kid-friendly on Saturday" or "a calm evening for two under 2,000 TL" are hard to express. Seat maps are flat diagrams, so buyers cannot judge how close or central a seat is until they arrive. During payment, seats can disappear without warning because there is no visible hold, and every user sees the same home page regardless of city, taste or budget.

Why it is worth solving. Live events are an expensive and usually shared purchase: a bad seat or a failed payment spoils an evening for several people. Clear information and a safe checkout save time for audiences and reduce abandoned purchases for organisers. The problem is also a good fit for this course, because it combines a relational database, user accounts and sessions, server-side validation, concurrency (two people selecting the same seat at the same moment), responsive front-end design and integration with an external AI web API.

Who benefits. Audiences — students on a budget, couples, parents and sports fans — get faster discovery and confident seat choices. Venues and organisers get fewer abandoned checkouts and no double bookings, and administrators get a simple way to keep the catalogue up to date.

What makes it useful and interesting. Perde combines four ideas that are rarely offered together: (1) a 3D preview of the stage from the selected seat; (2) a 10-minute seat hold with a visible countdown, enforced by the database so that a seat is never sold twice; (3) personal recommendations based on a short onboarding survey; and (4) *Suflör*, an AI assistant named after the theatre prompter, which understands natural-language requests, recommends only events that really exist in the catalogue and can open the seat map with the best seats already selected. The interface follows Apple's design language and is localised in Turkish for its target market.

2 Target Users and User Roles

The application has three user roles. The AI assistant is a feature available to guests and members, not a separate role.

ROLE	WHO THE USER IS	WHAT THE USER EXPECTS	WHAT THE USER CAN DO
Guest	Anyone who opens the site without signing in, often just looking for ideas.	To browse freely and quickly, without being forced to create an account.	Browse the Explore page and categories; search and filter events; open event pages, seat maps and the 3D seat view; read reviews; ask the AI assistant. Must sign in to hold seats or buy.
Member	A registered customer, e.g. a university student on a budget, a couple planning an evening out, a parent looking for children's shows, or a football fan.	A safe and quick purchase, all tickets in one place, and suggestions that match their city, taste and budget.	Everything a guest can do, plus: sign up / sign in / sign out; complete the onboarding survey; save favourites; hold seats and pay; view tickets with QR codes; rate and review events; get personalised recommendations.

Administrator	Platform staff (in this project, the team members) who maintain the content.	To keep the catalogue correct and the community content appropriate, and to see how events are selling.	Sign in to a protected admin area; create, edit and cancel events, sessions and ticket prices; hide inappropriate reviews; view sales reports. Administrators never see full card details.

3 Functional Requirements

Requirements are grouped by priority. *High* requirements form the core purchase flow and the AI feature; *Medium* requirements improve discovery and engagement; *Low* requirements cover administration.

ID	FUNCTIONAL REQUIREMENT	PRIORITY
CORE FLOW		
FR-01	Users can sign up with name, e-mail and password, sign in and sign out. A duplicate e-mail is rejected with a message, and the session persists across page reloads until the user signs out.	HIGH
FR-02	Users can browse upcoming events on an Explore page and by category (concert, theatre, stand-up, cinema, sports, children) in the selected city. Past sessions are not shown.	HIGH
FR-03	Users can search events by event, artist or venue name and filter the results by date range, price range, city/district, venue and genre. Active filters are stored in the URL, so opening the same link shows the same results.	HIGH
FR-04	Users can open an event page that shows the description, venue, duration, age limit, all sessions with date and time, and ticket tiers with prices and availability.	HIGH
FR-05	Users can choose a session and select up to 10 seats on an interactive seat map. Sold and held seats are shown as unavailable and cannot be selected. A “best seats” button selects the best available adjacent seats for the requested number of people.	HIGH
FR-06	When a signed-in member continues to checkout, the selected seats are held for 10 minutes and a countdown is shown. Held seats cannot be taken by another user and are released automatically when the hold expires or the order is cancelled.	HIGH
FR-07	Members can pay with a demo card form that validates the card number (Luhn check), expiry date and CVC. On success the order is marked as paid and one ticket with a unique QR code is issued per seat; on failure the seats stay held and the user can try again.	HIGH
FR-08	Members can view their upcoming and past tickets under “My tickets”, each showing the event, session, venue, seat and QR code.	HIGH
FR-09	Users can ask the AI assistant (Suflör) for suggestions in natural language. The assistant replies with event cards taken only from the catalogue and, on request, opens the seat map of a suggested event with the best seats pre-selected.	HIGH
DISCOVERY & ENGAGEMENT		
FR-10	Users can preview a 3D view of the stage from any selected seat; a 2D seat map remains available on devices without WebGL.	MEDIUM
FR-11	Members can add events to and remove them from their favourites and see all favourites in one list.	MEDIUM
FR-12	Members can rate an event from 1 to 5 stars and write one review per event (non-sports events). Submitting again updates the existing review. The event page shows the average rating and the reviews; reviews by ticket holders show a “Has ticket” badge.	MEDIUM

FR-13	New members can complete an optional onboarding survey (city, categories, genres, budget, company). The Explore page then shows a personalised “For you” list based on the answers.	MEDIUM
FR-14	The Explore page shows a daily “Best sellers” chart that ranks events by tickets sold.	MEDIUM
ADMINISTRATION		
FR-15	Administrators can create, edit and cancel events, sessions and ticket tiers. Cancelled sessions disappear from the catalogue and cannot be booked.	LOW
FR-16	Administrators can hide a review, after which it is no longer shown or counted in the average rating, and can view a sales report (tickets sold and revenue per event) for a chosen date range.	LOW

4 Non-Functional Requirements

ID	QUALITY	REQUIREMENT AND HOW IT WILL BE VERIFIED
NFR-01	**Performance**	Explore, category and event pages reach Largest Contentful Paint in ≤ 2.5 s on a simulated 4G connection (Lighthouse). Search and filter results update in ≤ 500 ms for a catalogue of about 100 events and 200 sessions.
NFR-02	**Security & data protection**	Passwords are stored only as salted scrypt hashes. Sessions use httpOnly, SameSite cookies. Every write operation is validated on the server and all database access uses parameterised queries. Full card numbers and CVC codes are never stored or logged — only the card brand and last four digits.
NFR-03	**Reliability & data integrity**	A seat can never be sold twice: a unique database key on (event, session, seat) rejects a second hold. In a test where two browsers hold the same seat at the same moment, exactly one succeeds and the other sees a clear message. Errors are shown as friendly messages, never as raw stack traces.
NFR-04	**Usability & responsive design**	All pages work from 360 px to 1920 px wide without horizontal scrolling. In a usability test with 5 first-time users, at least 4 buy a ticket for a given event within 3 minutes without help.
NFR-05	**Accessibility**	Pages follow WCAG 2.1 AA: text contrast ≥ 4.5:1, every control reachable by keyboard with a visible focus state, labels on icon buttons, and animations reduced when “reduce motion” is on. Lighthouse accessibility score ≥ 90 on the main pages.
NFR-06	**Browser compatibility**	The application works in the latest two versions of Chrome, Safari, Firefox and Edge on desktop, and in Safari on iOS and Chrome on Android.
NFR-07	**AI quality & availability**	The assistant’s first words appear within 3 s. In an evaluation set of 30 typical requests, 100% of suggested events exist in the catalogue with correct dates and prices. If the AI service is unavailable, the chat shows a friendly message and search and filters keep working.

5 User Scenarios

S1 Buy two seats for a theatre play

ACTOR	Member
PRECONDITIONS	The member is signed in. The play has an upcoming session with at least two free adjacent seats.
MAIN FLOW	1. The member searches for “Zengin Mutfağı” and opens the event page. 2. The member chooses the session on Wed 9 Dec, 20:30, and the “Full price” tier. 3. On the seat map, the member taps “Best seats” for 2 people; the system selects two adjacent seats. The member opens the 3D view to check the view of the stage. 4. The member continues to checkout. The system holds both seats for 10 minutes and shows the countdown. 5. The member enters card details; the system validates them and confirms the payment. 6. The system marks the order as paid and issues two tickets with QR codes.
EXPECTED OUTCOME	Both tickets appear under “My tickets”, and the two seats are shown as sold to all other users. Alternative: if the 10 minutes run out before payment, the seats are released and the member returns to the seat map.
REQUIREMENTS	FR-03, FR-04, FR-05, FR-06, FR-07, FR-08, FR-10

S2 Get a recommendation from the AI assistant

ACTOR	Guest or member
PRECONDITIONS	The AI service is available. The catalogue contains events in the coming week.
MAIN FLOW	1. The user opens Suflör from the navigation bar and types: “A romantic evening for two this weekend, under 2,000 TL each.” 2. The system sends the request, together with today’s date, the selected city and the catalogue, to the AI model. 3. The assistant replies with a short explanation and two event cards from the catalogue (date, starting price, rating). 4. The user writes: “Two seats for the first one, please.” The assistant checks availability and opens the seat map with the two best adjacent seats selected.
EXPECTED OUTCOME	The user is on the seat page with two seats selected and can continue with checkout (scenario S1, step 4). No event that is missing from the catalogue is ever suggested.
REQUIREMENTS	FR-09, FR-05, FR-06

S3 Find a concert with filters and save it

ACTOR	Guest, then member
PRECONDITIONS	The catalogue contains concerts in Istanbul.
MAIN FLOW	1. The guest opens the “Concert” category and sets the filters: city Istanbul, this weekend, maximum price 1,500 TL. 2. The system lists only matching concerts and updates the URL with the chosen filters. 3. The guest taps the heart on one concert; the system explains that favourites need an account and offers to sign in. 4. The guest signs in, returns to the same page and taps the heart again; the concert is added to favourites.
EXPECTED OUTCOME	The favourites list contains the concert. Opening the copied URL in another browser shows the same filtered list.
REQUIREMENTS	FR-01, FR-02, FR-03, FR-11

S4 Rate a show after attending it

ACTOR	Member who bought a ticket
PRECONDITIONS	The member is signed in and has a paid ticket for a non-sports event.
MAIN FLOW	1. The member opens the event page and goes to “Ratings & reviews”. 2. The member chooses 4 stars, selects the tags “Acting” and “Script” and writes a short comment. 3. The system validates the review (rating 1–5, one review per member and event) and saves it.
EXPECTED OUTCOME	The review appears at the top of the list with a “Has ticket” badge, and the average rating is updated. Submitting again edits the same review instead of adding a second one.
REQUIREMENTS	FR-12, FR-08